Mohammad Ahmad

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TECHNICAL SKILLS

- **Languages:** Python, C/C++, C#, Java, JavaScript, TypeScript, HTML/CSS, VHDL, VBA, SQL
- **Frameworks & Tools:** React, Node.js, Power BI, Excel (Advanced), Office 365, SharePoint, Unity
- **Software:** MATLAB/Simulink, Vivado, AutoCAD, SolidWorks, Adobe Photoshop, Adobe Premiere Pro, Jira, VS Code
- **Hardware & Systems:** Arduino, Raspberry Pi (GPIO/I2C), ADC interfacing, Embedded Systems, PLCs, FPGA (Nexys A7), Digital Systems, Robotic Systems, PCB Design (KiCad), basic circuits
- **Protocols & Interfaces:** I2C, SPI, USB, TCP/IP, WebSockets, HTTP, JSON
- **Spoken Languages:** Fluent in English and Arabic

EDUCATION

Bachelor of Engineering (Computer Engineering Co-op)

University of Guelph, Guelph, ON September 2022 - Present | Expected Graduation: April 2027

EMPLOYMENT EXPERIENCE

Cooke & Denison Ltd. - IT Project Assistant (Co-op) **Guelph, ON | January 2025 - December 2025**

- Built a two-click, multi-file Excel quoting engine (VBA, SQL, Python) with type-specific machining-time models; auto-generates Word/PDF quotes, logs to a database and refreshes inventory, cutting quoting time from ~2 hours to ~5 mins
- Integrated CNC machine data with Microsoft Business Central/Shop Floor Insight and produced Power BI dashboards for uptime/downtime and order progress
- Automated office workflows (punch in/out reports, supplier purchase requests, ready-to-send quotes) and validated against historical estimates for accuracy and consistency

Youthforce - Team Supervisor **Kitchener, ON | July 2022 - September 2024**

- Led a team of seven individuals in the Region of Waterloo's Youthforce program, coordinating landscaping projects across municipal properties
- Redesigned the region's project proposal creation system using Excel to better communicate project vision and speed up the process by 40%, receiving strong feedback from regional management

PROJECTS EXPERIENCE

Raspberry Pi Digital Voltmeter **Guelph, ON | September 2025 - December 2025**

- Built a Raspberry Pi voltmeter reading 0-3V analog input via ADS1015 (12-bit ADC) over I2C and implemented multiplexed 3-digit 7-segment display using GPIO timing loops and transistor digit selection in low-level C (libgpiod, lgpio)
- Validated accuracy against oscilloscope/function generator measurements and logged samples to CSV for analysis

Matrix Multiplication Hardware Accelerator **Guelph, ON | September 2025 - December 2025**

- Designed and implemented a 3×3 matrix multiplication hardware accelerator in VHDL on the Nexys A7 FPGA (100 MHz) using Vivado, building an FSM-controlled iterative MAC datapath to compute nine dot-products deterministically
- Packed/unpacked matrix data using flattened bus interfaces (two 54-bit inputs, one 126-bit output) and implemented time-division multiplexed 7-segment display control to show selectable output elements via board switches

Adder Architecture Design & Performance Analysis **Guelph, ON | September 2025 - December 2025**

- Designed and implemented RCA, CSA, and CLA architectures in VHDL, parameterized for 16-, 32-, and 64-bit operation using generics and generate statements
- Performed post-implementation timing analysis in Vivado, analyzing critical paths, slack, and Fmax to demonstrate area/speed/power tradeoffs and why CLA meets 100 MHz at 64-bit while RCA fails due to carry propagation

MIPS Architecture **Guelph, ON | January 2025 - April 2025**

- Designed and implemented a MIPS architecture computer in VHDL, developing an ALU, Control Unit, and Register File resulting in a functional processor that executed instructions correctly
- Developed and tested instruction fetch, decode, execute, and memory access pipeline stages using VHDL simulations to verify processor behavior and instruction execution

Guelph Engineering Competition **Guelph, ON | October 2024**

- Collaborated with three peers to design, build, and program an autonomous robot using Arduino/Python for a competitive robotics challenge, implementing sensor integration (ultrasonic, line-following) and mechanical systems (gripper mechanisms, motor control)
- Achieved second place with minimal point gap from the first place team